Recent Advancements of Plant-Based Natural Fiber–Reinforced Composites and Their Applications

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Abstract

Demands for reducing energy consumption and environmental impacts are the major driving factors for the development of natural fiber-reinforced composites (NFRCs) in many sectors. Compared with synthesized fiber, natural fiber provides several advantages in terms of biodegradability, light weight, low price, life-cycle superiority, and satisfactory mechanical properties. However, the inherent features of plant-based natural fibers have presented challenges to the development and application of NFRCs, such as variable fiber quality, limited mechanical properties, water absorption, low thermal stability, incompatibility with hydrophobic matrices, and propensity to agglomeration. Substantial research has recently been conducted to address these challenges for improved performance of NFRCs and their applications. This article reviews the recent advancements of plant-based NFRCs, focusing on strategies and breakthroughs in enhancing the NFRCs' performance, including fiber modification, fiber hybridization, lignocellulosic fillers incorporation, conventional processing techniques, additive manufacturing (3D printing), and new fiber source exploration. The sustainability of plant-based NFRCs using life-cycle assessment and the burgeoning applications of NFRCs with emphasis on the automotive industry are also discussed.

Keywords:

A. *Polymer-matrix composites (PMCs);*

A. *3D reinforcement;*

B. *Mechanical properties;*

B. *Fiber/matrix bond;*

1. Introduction

Natural fiber–reinforced composites (NFRCs) attract increasing interest as an alternative to metals and synthetic fiber-reinforced composites primarily because of the growing demands for light weight in materials and a reduced environmental footprint [1]. Currently, glass fibers are the dominant reinforcing fiber in the polymer matrix. In 2015, ~95% of the fiber composites were reinforced using glass fibers [2]. Compared with glass fiber, natural fibers have several advantages, such as lower density, biodegradability, abundant availability, good damping properties, less abrasive damage to equipment, and high health safety (i.e., low skin irritation), making them a promising alternative reinforcing material for composites (**Figure 1**). Additionally, life-cycle assessment (LCA) studies have reported that NFRCs are environmentally superior to glass fiber-reinforced composites [3]. The Natural Fiber Composites Market forecast report has projected that the global NFRCs market size is expected to reach $10.89 billion by 2024, from $4.46 billion in 2016 [4]. Growing demands for lightweight and fuel-efficient vehicles will further propel the growth of NFRCs in the market [5].

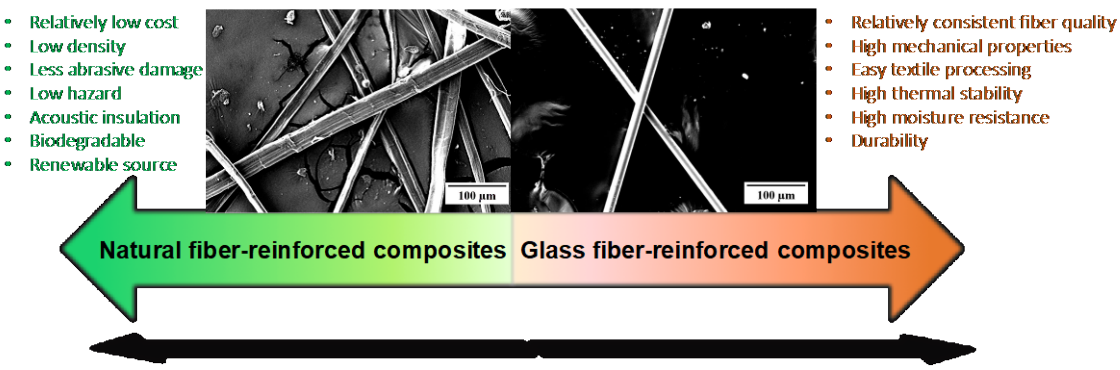

Figure 1. Relative comparison between natural fiber and glass fiber for composite manufacturing [6, 7]. Insets are SEM photos of (left) alkaline mercerized ramie fiber and (right) E-glass fiber.

The performance of plant-based NFRCs is determined by a synergic combination of factors derived from the properties of reinforcing fiber, base matrix, and the interfacial interaction of the fiber and matrix (e.g., mechanical interlocking, physical adhesion, chemical bonding). Despite the great potentials of natural fiber described previously, several innate natural fiber features have been considered as major technical barriers for the production of high-performance composites and their applications. These challenges include the heterogeneous characteristics of natural fiber (i.e., variation in the cell wall structure, composition, and geometry), leading to a wide variation of fiber quality; relatively lower mechanical properties; hydrophilicity, leading to incompatibility and aggregation tendency in hydrophobic polymer matrix; high water absorption; low thermal stability; and difficult processability into yarns and fabrics compared with glass fibers, impeding the manufacturing of high-performance NFRCs [8-10]. Some of these challenges of natural fibers have been substantially addressed by recent development in, for example, fiber treatment and modification, and process and product innovation. The improvement in the performance of NFRCs will continue to push forward the global demand for NFRCs and their applications.

In this review, we provide an overview of the recent advancements of plant-based NFRCs. Other naturally existing fibers including animal fibers (i.e., protein fibers such as silk and wool) and bacterial fibers are not covered in this review. We first briefly review the quality and mechanical properties of natural fibers from plants commonly used for manufacturing NFRCs. Then, mechanical properties of NFRCs made from three widely used thermoplastic polymer matrices—polypropylene (PP), polyethylene (PE), and polylactic acid (PLA) are discussed, with an emphasis on tensile properties. The strategies and breakthroughs in improving the performance of NFRCs are then discussed and highlighted, including fiber treatment and modification, fiber hybridization, incorporation of lignocellulosic fillers, advanced processing techniques, 3D printing, and exploration of new fiber resources. Finally, the life-cycle analyses and major applications of NFRCs are briefly discussed.

2. Natural fibers and their mechanical properties

The production of cellulose by plant photosynthesis is estimated to be 10^{12} tons per year in the biosphere, making polysaccharides the largest organic carbon reservoir [11]. The mostly studied cellulose-based natural fibers can be categorized depending on the plant species and plant tissue used [10, 12, 13], such as bast (e.g., banana, flax, hemp, jute, kenaf, ramie, rattan), leaf (e.g., abaca, agave, banana, cantala, coroa, curaua, henequen, istle, manila, phormium, piassava, pineapple, palm, sansevieria, sisal), seed and fruit (e.g., coconut, coir, cotton, kapok, rice husk), and stalk (e.g., bagasse, bamboo, barley, candillo, corn, isora, kudzu, nettle, oat, rape, reed, rice, rosselle, rye, wheat, wood). Plant-based natural fibers are usually extracted from the plant material after a retting or decortication process to remove the undesired cell wall components (e.g., pectin, extractives, hemicellulose, and lignin).

Several different stages are involved in manufacturing natural fibers, including plant growth, harvesting and processing, fiber isolation, and supplying [14]. Even from the same plant source, the properties of natural fiber are complicated by many factors, including physical features, chemical composition, crystalline cellulose dimensions, microfibrillar angle, defects, structure, and isolation method. Consequently, the quality and mechanical properties of the fiber can be substantially different. **Table 1** tabulates the properties of commonly studied natural fibers in comparison with several select synthetic fibers. The tensile strengths and moduli of these natural fibers cover a broad span of values with the typical tensile strength in the range of 100 to 1,000 MPa. The tensile strengths of natural fiber are generally lower than some listed synthetic fibers (e.g., the tensile strength of E-glass fiber is about 2,000–3,500 MPa). This inferiority also applies to the elastic moduli of natural fibers compared with glass fibers. However, the specific moduli of several natural fibers (e.g., flax, hemp, kenaf, and wood pulp) are similar or even superior to certain synthetic glass fibers when the fiber density (1.2–1.6 g/cm^3 for natural fiber vs. 2.5–2.6 g/cm^3 for glass fiber) is considered.

(Table 1. Here)

3. Mechanical properties of NFRCs

Thermoplastic and thermoset polymers have both been used as matrices for natural fibers. Because most natural fibers are not thermally stable above 200°C, thermoplastics that soften and thermosets that can be cured below this temperature are in turn selected as a matrix [22]. Due to the easy recycling potential of thermoplastics, four types of matrices (i.e., PP, low-density PE [LDPE], high-density PE [HDPE], and PLA) have been widely used for NFRCs. The major chain structure, physical features, and degradability of the four matrices are compared in Table 2. The mechanical properties of their fiber–reinforced composites are discussed separately in the following sections.

(Table 2 here)

3.1 Natural fiber–reinforced PP

PP is widely used as a matrix for NFRCs because of its processing suitability (i.e., filling, reinforcing, and blending), affordability, low heat distortion temperature, transparency, flame resistance, and dimensional stability [26]. Although the tensile properties of natural fiber/PP composites exhibit a wide distribution, incorporating natural fibers can substantially increase the elastic modulus and the tensile strength of the matrix depending on the fiber content and incorporation methods (**Figure 2**). Compared with the tensile properties of 20 wt % long glass fiber–reinforced PP (orange dotted line in **Figure 2**) that is specified for interior applications from Ford Motor

Company [27], many NFRCs with fiber content higher than 30 wt % have attained higher tensile modulus values. However, the tensile strengths of NFRCs are typically lower than this glass fiber/PP composite, which is historically attributed to the incompatibility between natural fiber and the matrix. This lower tensile strength could be also associated with the low transverse mechanical properties of natural fiber relative to glass fiber.

Interestingly, bamboo fiber has shown higher physical and chemical compatibility with PP than glass fiber as revealed by the better combination of wetting parameters (e.g., higher work of adhesion, higher spreading coefficient, higher wetting tension, and lower interfacial energy) [28]. However, the glass fiber exhibited higher interfacial bonding strength in the composites, indicated by both higher critical local interfacial shear stress value and radial normal stress at the moment of crack initiation, which explained that the overall performance of bamboo fiber was weaker than glass fiber in reinforcing PP. This result supports the idea that the anisotropic nature of natural fibers is likely the primary reason for its lower interfacial properties and stress transfer capability at the fiber/matrix interface compared with glass fiber. Thomason reported that the transverse and shear modulus of jute fiber is an order of magnitude lower than the longitudinal modulus [29]. Therefore, the general lower mechanical performance of NFRC to the glass fiber–reinforced composites analog is primarily attributed to the low mechanical properties in the transverse direction with the potential detachment of the outer layer and elementary fibers in contact with the matrix, often the case in compounding and extrusion, rather than the physical and chemical compatibility between the fiber and matrix.

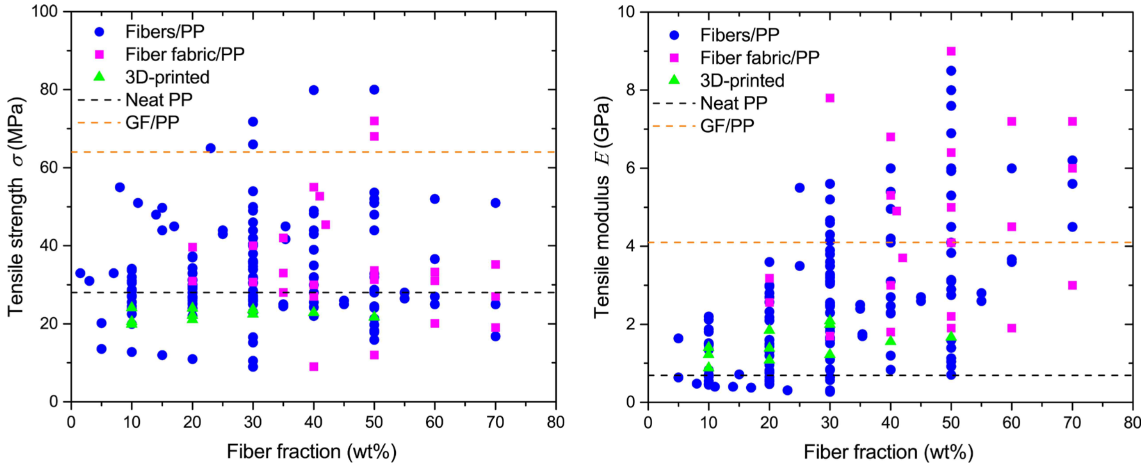

Figure 2. Comparison of the reported tensile strengths (σ) and moduli (E) of natural fiber–reinforced PP. Volume fiber fraction is used if the weight loading fraction is not reported. The black dotted line represents the properties of neat PP (σ of 28 MPa and E of 0.69 GPa) [30] and the orange dotted line represents the properties of 20 wt % long glass fiber (GF)–reinforced PP specified for interior applications from Ford Motor Company (σ of 64 MPa and E of 4.1 GPa) [27]. The complete numbers with each reference are included in the supplementary file.

3.2 Natural fiber–reinforced PE

PE is generally divided into LDPE and HDPE based on its density and branching. LDPE has relatively long branches with a random packing pattern and low crystallinity, whereas HDPE consists of long chains (without major branching) with a regular packing pattern and high crystallinity [31]. Similar to natural fiber/PP composites, the tensile properties of natural fiber/PE composites exhibited a wide distribution, but the tensile strength and elastic modulus can be both substantially reinforced by natural fibers (**Figure 3**). Compared with the 20 wt % glass fiber–reinforced HDPE (orange dotted line in **Figure 3**) [32], higher stiffness and comparable tensile strength of NFRCs can be achieved. In particular, the NFRC composites fabricated from natural fibers in mat or stacking sheets demonstrated promising potential as an alternative to glass fiber–reinforced composites [33]. For example, the HDPE composite with 40 wt % kenaf fiber mat showed a specific tensile strength of 50 MPa·cm^3/g, tensile modulus of 6.3 GPa·cm^3/g, flexural strength of 72.9 MPa·cm^3/g, and flexural modulus of

3.9 GPa·cm^3/g, which are comparable to the 40 wt % discontinuous glass fiber/HDPE counterpart with 47 MPa·cm^3/g, 6.6 GPa·cm^3/g, 66.1 MPa·cm^3/g, and 4.4 GPa·cm^3/g, respectively. A similar process combining an adhesive coating of flax fabric, sandwiching coated fabric with polymer films, and compression molding of multilayer prepregs obtained flax/PE composite with exceptionally impressive tensile strength of 180–280 MPa and tensile modulus of 15–22 GPa [34].

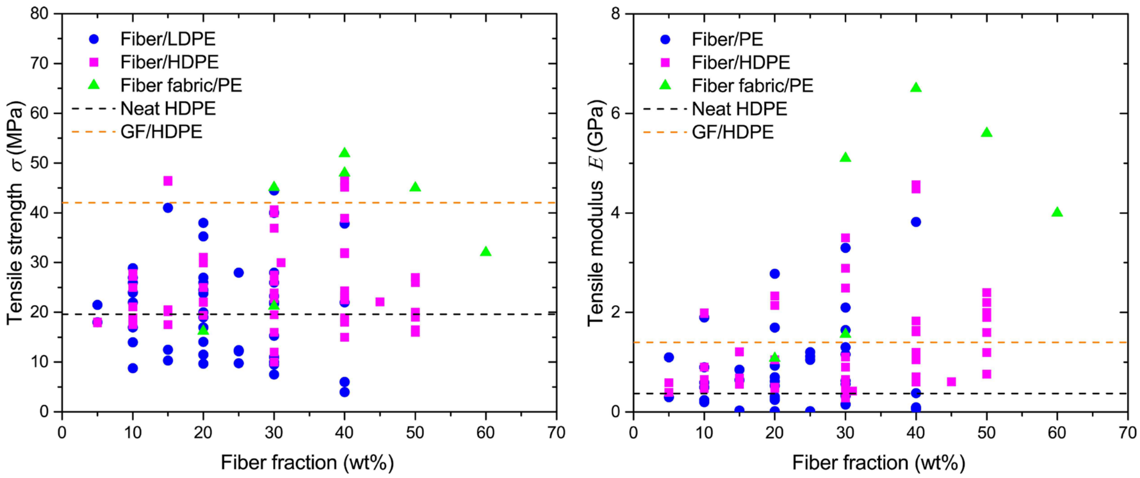

Figure 3. Comparison of the reported tensile strengths (σ) and moduli (E) of natural fiber–reinforced PE. Volume fiber fraction is used if the weight loading fraction is not reported. The black dotted line represents the properties of a neat HDPE (σ of 19.6 MPa and E of 0.74 GPa) [35] and the orange dotted line represents the properties of 20 wt % glass fiber (GF)–reinforced HDPE (σ of 42 MPa and E of 1.4 GPa) [32]. The complete numbers with each reference are included in the supplementary file.

3.3 Natural fiber–reinforced PLA

PLA has emerged as an interesting matrix with outstanding physical and mechanical properties, renewability, and biodegradability. Additionally, PLA has special qualities including good transparency and processability, glossy appearance, and high rigidity for NFRC manufacturing, although it has some shortcomings such as low thermal stability, brittleness, and high rate of crystallization [36]. Compared with the 20 wt % glass fiber–reinforced PLA [37], many NFRCs have achieved higher tensile strengths and moduli (**Figure 4**). The mechanical properties of PLA composites can be significantly enhanced by controlling the fiber content, fiber length [38], fiber direction [39], prepreg form [40], and woven structure [41]. For example, the composite made from wood pulp sheets compressed with PLA film attained significantly improved tensile properties (σ of 121 MPa and E of 10.5 GPa) to the neat PLA (σ of 60 MPa and E of 3.5 GPa) [40]. This wet-laid paper-making method showed several advantages including uniform dispersion of fiber, minimal reduction of fiber aspect ratio and mechanical damage, and high fiber fraction.

The tensile properties of PLA composites can also be significantly improved when incorporating unidirectional natural fibers. For example, PLA with 44 vol % unidirectional flax had a specific tensile strength of 252.3 MPa·cm^3/g and modulus of 14.9 GPa·cm^3/g in the fiber direction that were even slightly superior to the 43–50 vol % woven glass fiber–reinforced epoxy [39]. The unidirectional nature of natural fiber maintained great reinforcement in the composite strength and modulus that were even better than PLA composites made with fibers in mat layers and injection molded short fiber composites. The measured fiber orientation factor of PLA composites made from aligned discontinuous harakeke and hemp fiber showed higher values than the composites prepared using injection and compression with randomly oriented fiber mats, but slightly lower than those obtained with aligned fiber nonwoven preform composites [42]. When fibers are aligned parallel to the loading direction with high interfacial bonding, the stress transfer from the matrix to fiber becomes more effective, thus leading the composite to a larger strength and stiffness. However, too much fiber content in the mat can reduce fiber orientation factor due to a higher degree of fiber agglomeration. Additionally, compared with glass fiber, natural fiber also tends to bend and twist in the matrix during extrusion and molding process, which could adversely influence the toughness and strengths of composites [43].

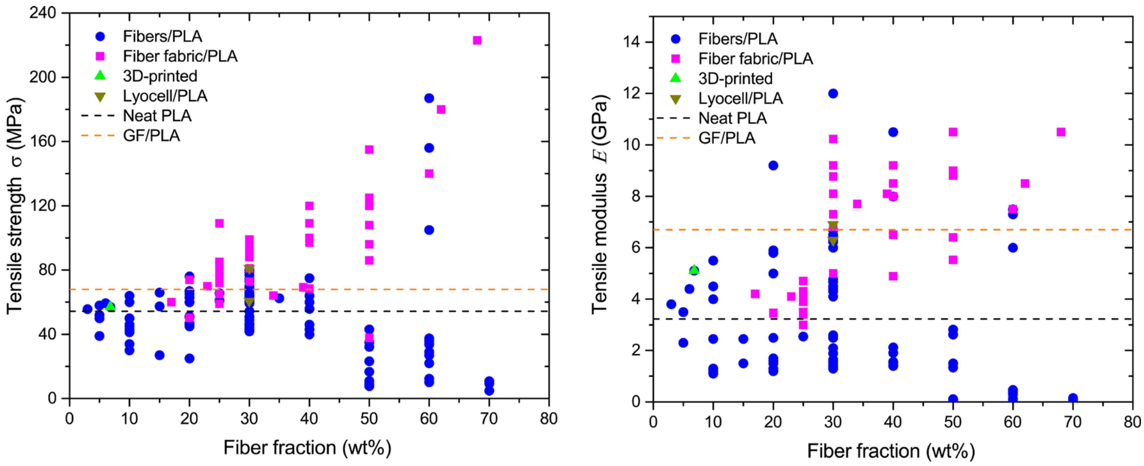

Figure 4. Comparison of the reported tensile strengths (σ) and moduli (E) of natural fiber–reinforced PLA. Volume fiber fraction is used if the weight loading fraction is not reported. The black dotted line represents the properties of neat PLA (σ of 54.36 MPa and E of 3.23 GPa) [37] and the orange dotted line represents the properties of 20 wt % glass fiber (GF)–reinforced PLA (σ of 67.9 MPa and E of 6.7 GPa) [44]. The complete numbers with each reference are included in the supplementary file.

Although certain NFRCs have achieved performance comparable to glass fiber–reinforced composites [33, 34, 39], NFRCs still exhibit several detrimental problems that have impeded their widespread applications. The challenges associated with NFRCs include (1) wide distribution of mechanical performance due to the heterogeneous physicochemical nature of natural fiber, various fiber isolation, and composite fabrication methods; (2) relatively low mechanical properties and durability restricting NFRC applications to nonstructural and interior components; (3) incompatibility of hydrophilic fiber with primarily hydrophobic matrices; (4) high water absorption of natural fiber causing comprised mechanical properties; (5) difficult dispersion of natural fiber in polymer matrix due to aggregation tendency and physical entanglement; and (6) low thermal stability resulting in restriction in the temperature range of manufacturing, processing, and final applications.

4. Strategies for enhancing the performance of NFRCs

Most NFRCs using the three common polymer matrix have maximum tensile strengths and moduli in the range of 20–140 MPa and 1–10 GPa (**Figures 2, 3, and 4**), respectively, which are low for primary and load-bearing components. This lower mechanical performance of NFRCs derives from several factors associated with NFRCs, such as heterogeneous characteristics of natural fiber leading to a wide variation of fiber quality; hydrophilicity leading to incompatibility and aggregation tendency in hydrophobic polymer matrix, high water absorption, low thermal stability; and difficult processability into yarns and fabrics compared with glass fibers. Several strategies, such as fiber treatment and modification, hybridization and structural configuration, incorporation of nanocellulose, manufacturing process advancement, 3D printing, and new plant fiber exploration have been used to negate these disadvantages.

4.1 Fiber treatment and modification

The incompatibility and poor adhesion of natural fiber in a polymer matrix are usually addressed by fiber treatment and modification to enhance effective wetting and uniform dispersion (**Figure 5**). The primary techniques used for fiber treatment and modification can be grouped into fiber pretreatment (e.g., mercerization), surface coating modified with coupling agents, and in situ compatibilization during processing depending on the practical applications. Mercerization, a chemical treatment using alkali, is widely used to fibrillate and purify fibers (partially removing oil, wax, pectin, hemicellulose, and lignin) prior to composite fabrication [45-48]. Fiber treatment with alkali has shown at least four favorable effects to the mechanical properties improvement of NFRCs: (1) increased cellulose content and the degree of crystallinity, which is indicative of higher fiber strength and stiffness; (2) increased surface roughness topography for better mechanical interlocking between the fiber and matrix; (3) increased cellulose exposure for increased bonding/reaction sites on the fiber surface; and (4) increased surface energy for better wetting and compatibility. Treatment with a mild alkaline condition (~5 wt %) is typically sufficient to remove fiber impurities with minimal impact on the fiber texture and structure whereas higher alkaline concentration can lead to excessive removal of lignin and fiber damage [46, 49-52].

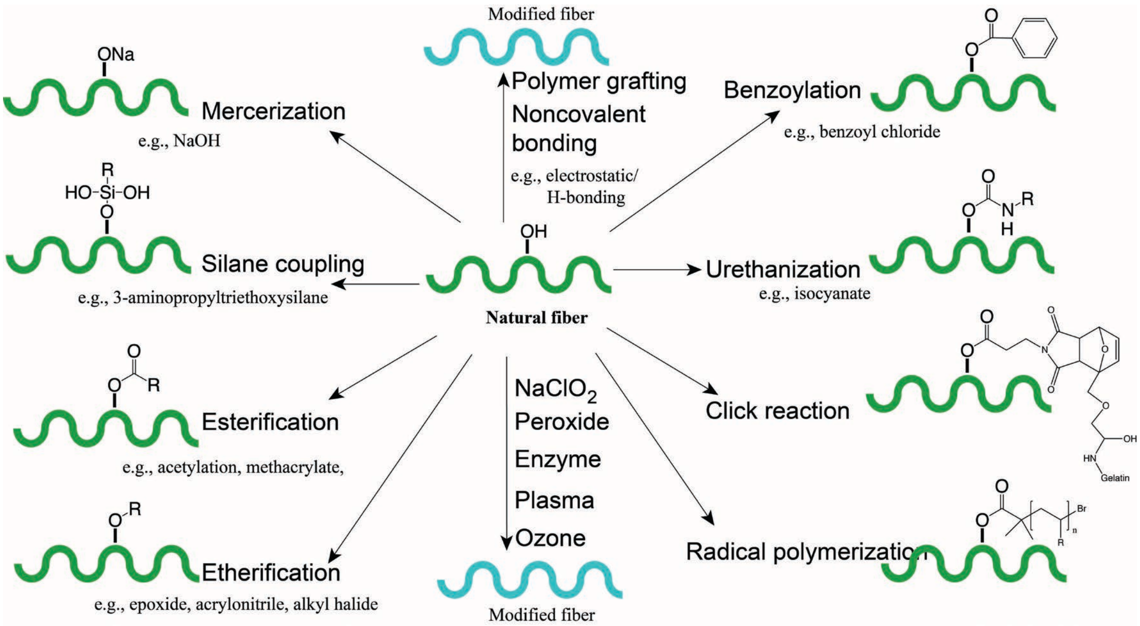

Figure 5. Examples of typically reported methods for treating and modifying natural fibers [53, 54].

Incorporating a coupling agent (e.g., maleic anhydride [MA]–grafted copolymer and silane agent) is another approach to improve the overall quality of NFRCs by enhancing interfacial bonding [55-58]. The anhydride groups of the MA-grafted copolymers can esterify with the surface hydroxyl groups of natural fiber and reduce hydrogen bond formation. The other part of the copolymer with similar polarity entangles with the polymer matrix for improved compatibility. A recent study showed that the coupling agent not only affected the nucleation process but also increased crystallization kinetics [59]. The isothermal crystallization studies of the composites revealed different crystallization kinetics of miscanthus/PP with and without the addition of MA-grafted PP (MAPP) in that the crystallization started earlier with larger values of the kinetic constant for the NFRCs containing MAPP. The study also suggested that the fiber size affects the crystallization process of the composite only when the MAPP is added—the rate constant of the PP crystallization increases gradually as the size of the miscanthus stem fragments is reduced. The increased surface area associated with fiber size reduction is linked with the increased nucleating capacity of fiber fragments acting as a nucleating agent. For non-coupled composites, the size of fiber fragments did not significantly influence the crystallization kinetics of miscanthus/PP. MA-grafted PLA is not as readily available as MAPP. However, its addition has shown improved interfacial adhesion for kenaf, wheat straw, coconut fiber, ramie, wood fiber, and reishi (*Ganoderma lucidum*) fiber in PLA composites [60-62].

El-Sabbagh performed a systematic work of different parameters, including coupling agent weight ratio to the fiber, coupling agent source, fiber type, and fiber weight content, to optimize the MAPP amount in natural fiber/PP composites [63]. The optimal stiffness, tensile, and impact strengths varied depending on the coupling agent loading as well as the fiber type, fiber loading, and coupling agent source. The MAPP to fiber ratio required for maximizing stiffness was less than that needed for tensile and impact strength. Taken together, 6.7 wt % of the coupling agent to fiber was found sufficient for significant improvement in mechanical properties. However, excessive MAPP did not increase the coupling effect with fibers but instead led to self-reaction and weak layers. Additionally, the interfacial adhesion and mechanical and thermal stability of the MAPP-coupled composites strongly depended on the amount of MA-grafting level and the MAPP molecular weight [64]: (1) a low MA-grafting level did not offer sufficient interaction with natural fiber whereas an excessively high MA percentage may restrict the interaction of MAPP with the matrix; and (2) MAPP with low molecular weight entangled insufficiently with the matrix whereas excessively high molecular weights of MAPP restricted its residence at the MAPP/matrix interface.

Silane coupling agents are another “family” of compounds commonly applied to fibers to increase the interfacial bonding of the fiber and matrix [53, 65]. The chemical modification of fibers with a silane agent undergoes the following major steps [65]: (1) hydrolysis of the silane agent to liberate reactive silanol group in the presence of water; (2) self-condensation of silanols (this step should be minimized usually by controlling pH); (3) physical absorption of reactive silanol compound to hydroxyl groups of the fiber through hydrogen bonding; and (4) grafting to form covalent –Si–O–C– bonds between silanol and hydroxyl groups upon heating. The functions of different silane treatment parameters, such as the concentration of the organosilane solution, soaking time, and treatment temperature, have been studied on flax/PLA composites. Flax fiber treated with 5–10 wt % (3-glycidyloxypropyl) trimethoxysilane at 70°C was recommended for the optimum organosilane treatment conditions, resulting in 20–25% higher yield stress than the composite without fiber treatment [66]. Treatment with γ-methacryloxypropyltrimethoxysilane had greater efficiency in hemp fiber fragmentation and a greater improvement in the tensile modulus of the PP/hemp composites than the other two silane agents, (γ-aminopropyltriethoxysilane, γ-glycidoxypropyltrimethoxysilane) [67]. The superior treating effect of γ-methacryloxypropyltrimethoxysilane was attributed to more stable adsorption and interaction to hemp fiber and less lubricating and self-condensation.

Fiber/matrix adhesion has been advanced through interfacial topology analysis of single-fiber composites. The interfacial shear strength of single sisal fiber/PP composite was found to be closely correlated with three types of crystalline interfacial crystalline morphologies: spherulite, medium–nuclei density transcrystallinity (MD-TC) and high–nuclei density transcrystallinity (HD-TC) (**Figure 6**) [68]. The transcrystalline superstructure depicts a columnar crystalline layer covering the fiber surface, and its growth varies on the nucleus density on the surface of the fiber. At low nucleus density, the crystals grew in both parallel and perpendicular directions to the fiber axis, namely MD-TC; at high nucleus density due to the presence of MAPP and shear-induced pulling, the crystal grew perpendicularly to the fiber axis, namely HD-TC. Compared with the amorphous conditions, the transcrystalline superstructure showed significantly higher interfacial shear strength, 95% for MD-TC and 178% for HD-TC. This increase was attributed to the interphase thermal shrinkage, leading to a high contraction force that can distribute the stress down to the atomic scale to affect the mechanical interlocking.

To promote the epitaxial growth of this transcrystalline structure, the addition of a rare earth compound into sisal/PP and cotton/PP systems triggered the rare earth compound adsorption onto the surface of natural fiber due to hydrogen bonding between the amide groups of the rare earth compound and the hydroxyl groups of fiber [69, 70]. The introduced particle generated a root-like fiber with an interfacial interlocking effect that significantly increased the interfacial adhesion, leading to higher interfacial shear strength, interfacial friction, and debonding energy [69]. Similarly, in another study, coating with few-layer graphene oxide nanosheets to ramie fiber showed significantly increased shear strengths in PLA [71]. Coverage of graphene nanosheets on ramie through hydrogen bonding exhibited a prominent capability to trigger the generation of highly ordered transcrystallinity of PLA composite with a higher orientation degree. These findings suggest that manipulating the formation of transcrystalline structures accompanied by enhanced interfacial shear strength may be an effective method for improving the mechanical properties of NFRCs. However, a more recent study has shown that coating wood fiber in PP with nucleating agents modified the matrix crystallinity and the transcrystallinity of the fiber and matrix, but had little influence on the mechanical properties composites [72]. This suggests the addition of nucleating agents likely has little influence on interfacial adhesion.

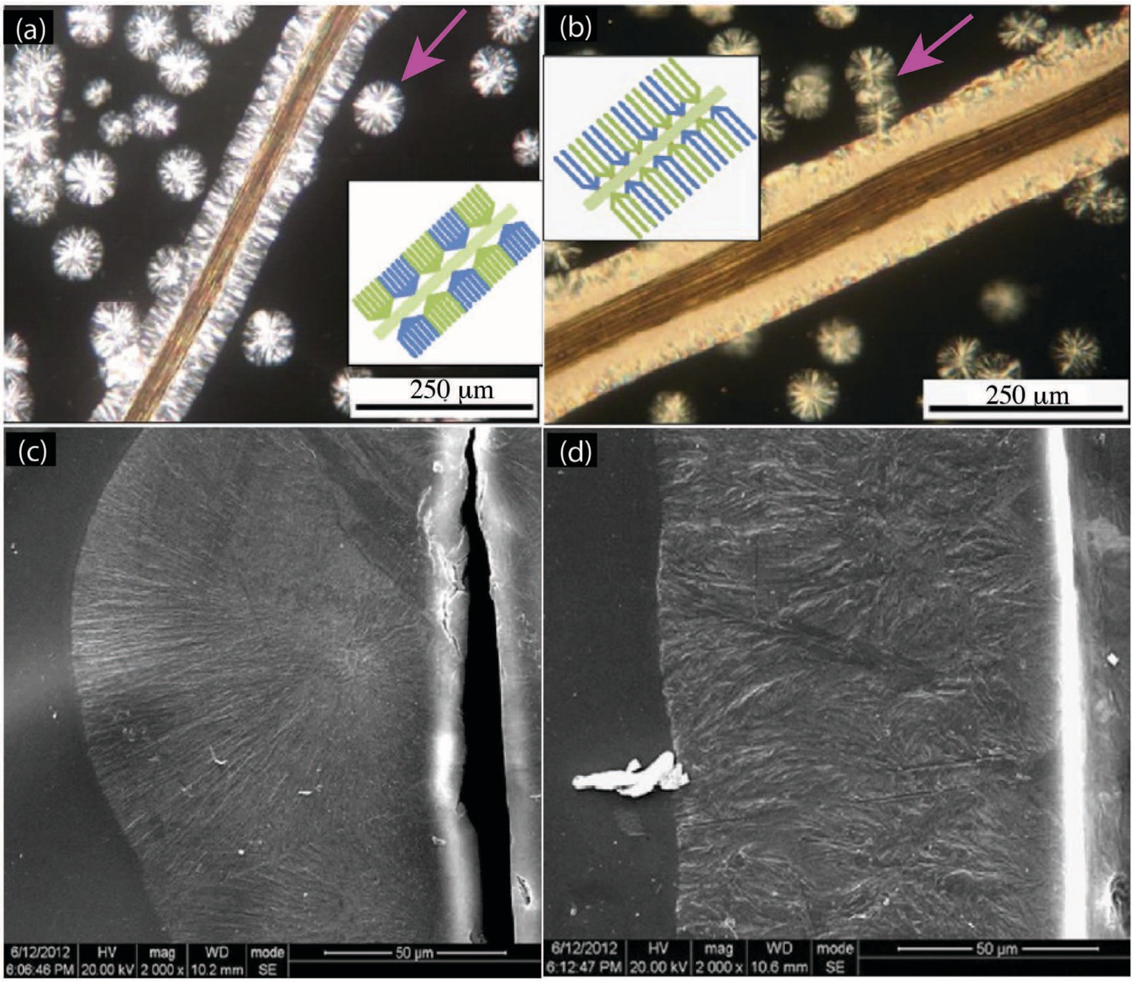

Figure 6. Micrographs revealing transcrystalline structure in a sisal fiber/PP composite. (a and b) Polarized optical images and (c and d) SEM images show three types of crystalline structures: spherulite (arrows in a and b), MD-TC (a and c), and HD-TC (b and d). Picture adapted from [68] with permission from Wiley.

4.2 Fiber hybridization

Fiber hybridization has been attempted to balance the deficiency of one specific fiber to achieve sustainability, low cost, and improved performance of NFRCs [73, 74].

Hybridization methods provide flexibility in fibers selection for engineering materials properties according to the application requirements. Fiber hybrid composites are typically produced by hybridizing natural fibers with another either natural or synthetic fiber(s) with superior properties, such as mechanical strength, chemical stability, nontoxicity, noncombustible, resistance to high temperatures, and thermal or acoustic insulation. Fiber hybridization is usually accomplished either via intermingling of different types of fibers as a mixture or interlaminating layers of different fibers before their addition to the matrix [75].

Hybridization of natural fibers with synthetic fibers, such as glass, carbon, Kevlar, basalt fibers, and carbon nanotubes, can significantly enhance the mechanical performance of NFRCs. For example, the hybridization of jute fabric with glass fiber fabric (1:1 weight ratio) has improved the tensile and impact strengths jute fabric-reinforced polyester by three-fold and six-fold, respectively; similarly, almost ten-fold of tensile strength and five-fold of the impact strength improvement is obtained by hybridizing jute fabric with carbon fibers [76]. To reduce the composite discrepancies due to shrinkage and warpage, the flow and cross-flow shrinkage values of the sisal/glass fiber/PP hybrid composite can be optimized by controlling the injection molding process parameters, thus meeting the material maximum shrinkage requirements for under-the-hood parts used in the automotive industry [77]. Hybrid composites of natural fiber with other natural fibers may not show comparable mechanical improvements to those hybridized with synthetic fibers, but this approach may have significant value in terms of sustainability. For example, a mixture of pineapple leaf fiber and fibers extracted from recycled bamboo chopsticks at 30 wt % improved the tensile strength, modulus, flexural strength, and modulus of neat PLA by

66.1%, 216%, 66%, and 121%, respectively [78]. This hybridization with PLA and poly(butylene succinate) can lead to fully biodegradable composites.

The ultimate performance of the hybrid composite is determined by several factors, such as fiber content, hybrid ratio, fiber orientation [79, 80], stacking sequences [76, 81-83], and innate fiber features. The tensile strength and failure strain of the hybrid composite increased from ~200 MPa and 0.9% for flax-alone reinforced composites to ~700 MPa and 1.4% for glass fiber–alone reinforced composites [84]. With a higher glass fiber hybrid volume fraction, the hybrid composite maintained its integrity until the failure dominated by the bigger failure elongation of glass fiber. Interestingly, the hybrid composite had a higher interlaminar shear strength compared with the composites reinforced with either natural or glass fiber alone. This was explained by an increase in interlaminar fracture toughness from the twisted flax yarn structures and rough surface that led to the bridging and entangling of interfaces.

Hybridization can compensate for the inherent disadvantages of the individual bioresources. For example, Yusoff et al. observed that the different stacking configurations of the hybrid PLA composite with kenaf(K)/bamboo(B)/coir(C) showed a diverse range of mechanical properties results [85]. With the same total fiber content (60 wt %), the stacking sequence of KBCCBK increased the neat resin tensile strength from 25 to 187 MPa and the modulus from 1.7 to 7.5 GPa. The increased tensile strength was about 20 and 78% higher than composite sequences of KCCK and BCCB, whereas the modulus was only 2 and 25% higher than those of KCCK and BCCB, respectively. The high Young's modulus of kenaf and bamboo fibers contributed better in resisting loads at the outer layers of composites, while the high ductile coir fiber offset the low elongation of the kenaf and bamboo fibers in the composites. In terms of the flexural properties, the PLA/BCCB and PLA/KBCCBK composites had about 20%

higher flexural strength but 70% lower flexural modulus than the PLA/KCCK composite. This higher flexural modulus of the PLA/KCCK composite was explained by the lumen connected to the secondary layer of the kenaf and coir fibers, which promoted cohesion by interlocking the fiber cell walls with the PLA matrix. The results suggest that high-strength and -modulus fibers in the outer layers and effective matrix impregnation into lumen are key factors for hybridization configuration maintaining superior mechanical performance. A recent study of the fiber aspect ratios in kenaf/corn husk for hybrid composites pointed out that hybrid fillers with similar dimensions (i.e., the aspect ratios of the reinforcing fibers are close to each other) were effective in external force transfer and modulus enhancement, which could be an indicator in optimizing the mechanical properties of a hybrid composite [86]. Also, Kwon et al. found that the aspect ratio of fibers before the extrusion process can be used directly to predict the mechanical properties because the aspect ratio after extrusion does not show a significant difference in the tensile modulus calculation.

4.3 Incorporation of nanocellulosic fillers

Recently, lignocellulosic nanofillers, such as cellulose nanocrystals (CNCs) and cellulose nanofibrils (CNFs), have emerged as attractive reinforcing materials due to their substantial specific surface areas, high crystal stiffness, strength, and maintained aspect ratio [87-89]. The crystalline cellulose was reported to have a tensile strength of 7.5 GPa and modulus of 110–220 GPa [19]. Similar to inorganic fillers, the micro- and nano-scaled fillers provide a large surface for linking and bonding the matrix, which is beneficial for physical adhesion at the interface, stress transfer, and ultimate mechanical properties of the composite [90]. **Table 3** lists select CNC- and CNF-reinforced composites and their essential properties. However, incompatibility and aggregation of nanocellulose in matrix remain the top challenges for fabricating high-performance nanocellulose composites.

(Table 3. Here)

With nanometric fillers, superior mechanical, thermal, and barrier properties can be achieved at low reinforcement concentration, usually <10 wt %. For example, softwood CNF–reinforced epoxy resin showed maximum values of tensile strength and modulus, impact strength, flexural strength and modulus, and elongation at 0.75 wt % fiber loading [97]. Reinforcement with a high concentration of nanocellulose fillers adversely leads to several problems, such as aggregation of nanofibers, reduction in fiber/matrix interfacial adhesion, restricted consolidation of composites during the crosslinking process, defects generation in the composites, ineffective stress transfer, and ultimately, compromised composite performance. Surface treatment of nanocellulose (e.g., silane agent, alkenyl succinic anhydride) is an applicable method to moderate nanofiber agglomeration and improve interfacial adhesion [93, 94, 98]. In theory, a well-dispersed mixture of 40 wt % CNF in cellulose acetate butyrate could increase the tensile modulus of the matrix from 2.07 GPa to 11 GPa [99]. The aspect ratio of nanocellulose is also critical for the high performance of nanocellulose-reinforced composites. By comparing the reinforcing effects of CNCs with different sizes isolated from the same biomass, high–aspect ratio CNCs were recommended not only because they decrease the percolation threshold and reinforcing capability at low CNC content, but also because the upper modulus limit corresponding to high CNC content was higher [101].

Another study of nanocomposites comparing CNF- and CNC-reinforced polyurethane revealed that CNFs with higher aspect ratio had better mechanical and thermal property improvement than CNCs [102]. Benhamou et al. also found that the residue of hemicellulose and lignin in CNFs resulted in better interfacial compatibility with the polymer. Similarly, Graupner et al. showed that lignin doping was beneficial for bonding and apparent interfacial shear strength of Lyocell fiber in PLA, MAPP, and PP matrices [103]. These benefits were generally attributed to several factors, including (1) the addition of lignin reduced the viscosity of PLA matrix; (2) exfoliated lignin particles led to a higher specific fiber surface for interfacial bonding; and (3) chemical interactions were improved between the fiber and matrix via the van der Waals force.

4.4 Advancement of conventional manufacturing processes

Extrusion, injection molding, compression molding, sheet stacking, and resin transfer molding are the most commonly used methods for manufacturing NFRCs. Process parameters such as compounding time, molding temperature, pressure, and compression time remain a complex balance but have a significant impact on the mechanical properties of NFRCs, such as wood flour/PP [104], flax/PLA [105], bamboo/epoxy [106], and kenaf/HDPE [107]. Studies have suggested that (1) wetting and homogeneous dispersion of fibers are the most crucial targets during NFRCs processing; (2) a relatively higher processing temperature and pressure generally have a better impact on rheological properties and penetration of polymer into the natural fibers, whereas the press/hold time has only a small impact; and (3) the thermo-chemical degradation of natural fibers should be considered as the maximum consolidation time to avoid major degradation of the composite mechanical strength. To reduce fiber attrition during processing, a direct-injection molding process (i.e., sisal and PLA pellets were directly fed into an injection-molding machine) showed significantly superior capability in maintaining fiber lengths compared with the extrusion-injection separate molding process [108]. However, the composites made from direct-injection molding had lower tensile and flexural properties than those of extrusion-injection molding, which was attributed to the formation of fiber clusters hindering adequate wetting of the fibers and acting as stress concentration under mechanical testing.

The effective dispersion of fibers is crucial for manufacturing high-performance NFRCs. A study of twin-screw extrusion of Cordenka cellulose/polyamide found that the extrusion temperatures and screw configurations caused only minor effects on the morphological structure and the mechanical properties of the composite and much more influence on fiber length distribution and dispersion [109]. Another study of kenaf/PP composites also noted that an effective fiber dispersion after the extrusion-injection process resulted in significantly improved mechanical performance irrespective of the initial fiber length [110]. The twin-screw extrusion of jute/PLA with a combined configuration that reduced the compounding shear stress and enhanced the distributive mixing greatly improved the overall fiber dispersion. This compounding process facilitated the dispersion of the jute yarn to bundle and the de-cohesion of the bundle to elementary fibers, leading to efficient strength transfer from matrix to fiber [111]. Short fiber–reinforcement was found to exhibit better PLA coating on the pullout fibers of the fractured composite compared with the long jute fiber–reinforcement (higher aspect ratio) (**Figure 7**). A recent study of injection-molded hemp/PP composites discovered that fiber content increased along the spiral flow length (i.e., longer fibers usually at the end of the injected part and shorter fibers near mold entrance point) [112]. The fiber content reached up to 30% deviation at different molding conditions, suggesting that the homogeneity of fiber weight content along the injection mold spiral acted as an essential quality feature [113]. Usually, a solid-state shear pulverization process is used to facilitate the dispersion and orientation of the filler particles within the polymeric matrix [91, 114].

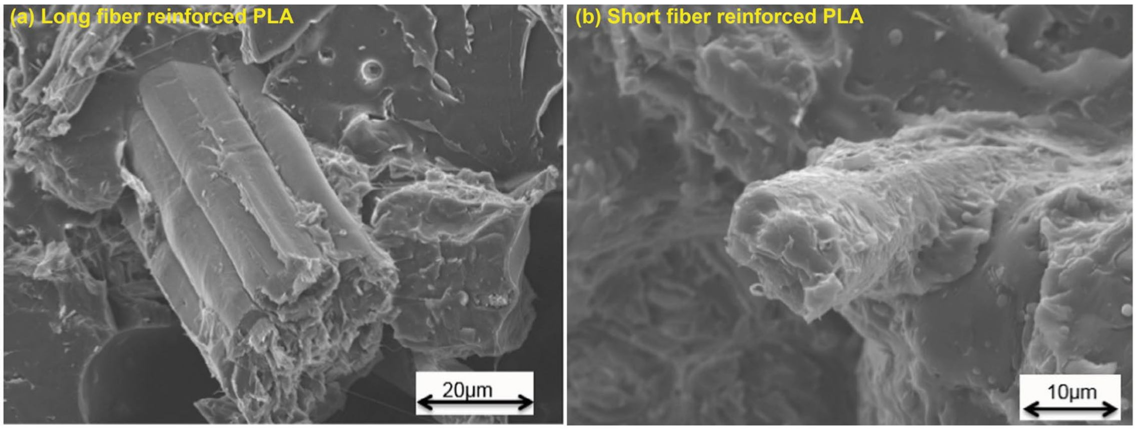

Figure 7. SEM micrographs of the fracture surface of composites reinforced with (a) long fiber pellets and (b) short fiber pellets. Reproduced from [111] with permission from Elsevier.

Recently, a new twin-screw processing method was developed that enables continuous micro-fibrillation of pulp and its melt compounding with PP [115]. In this method, never-dried pulp and PP powder were kneaded via a twin-screw extruder. During the melt compounding process, the water content was minimized by evaporation and the pulp was defibrillated and entangled in the powdered PP rather than the formation of microfibril cellulose aggregation. With the addition of 6.7 wt % MAPP, the PP composite with 50 wt % Kraft pulp had twice the tensile strength and 1.5 times the modulus of neat PP. The mechanical properties of PP composites were further improved when a cationic polymer coupling agent (i.e., vinyl amine-based polymer) was combined with MAPP [116]. The cationic polymer–bearing amino groups can react with MAPP, forming hydrogen bonds with cellulose. When a combination of MAPP and a cationic polymer was added, the PP composite with 30 wt % Kraft pulp had a 27% higher tensile strength and an 18% higher modulus than the composite with MAPP only.

4.5 Emergence of additive manufacturing of NFRCs

Additive manufacturing, or 3D printing, has recently been notably expanded to revolutionize the conventional manufacturing process across many industries. Compared with the traditional composite making processes mentioned previously, 3D printing has several advantages, including quick and easy manufacturing, reduced limitation on composites geometry, increased controllability of fiber volume and direction, reduced thermal degradation or deconstruction of natural fiber, and saving of expensive molds and tools. Considerable research has been conducted using forestry and agricultural fibers as 3D printing feedstock, such as microfibrillated cellulose and nanocellulose [117, 118]. Fused deposition modeling (FDM) is one of the simple and widely used 3D printing techniques because of the wide range of available polymers, such as PLA, polycaprolactone, and acrylonitrile butadiene styrene. In FDM, material filament consisting of fiber and matrix is guided through a heated nozzle with a feeding roller, and the melt paste is deposited layer by layer on a printing platform according to a computer-designed architecture and process (**Figure 8a**). The partially liquefied layer is then fused with the previous layer when the material solidifies during cooling.

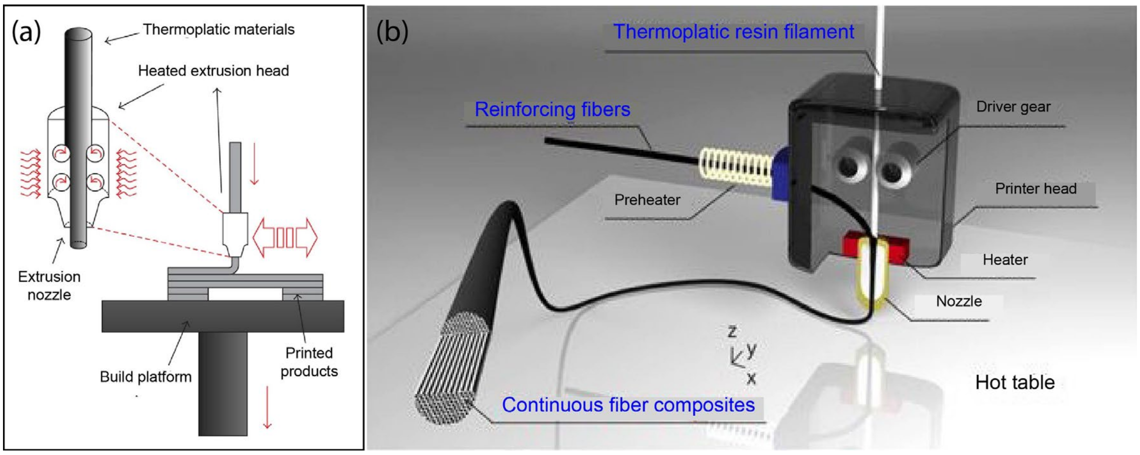

Figure 8. Schematic of FDM process (left) [119] and the FDM-based 3D printer head used to produce continuous fiber–reinforced composite using in-nozzle impregnation (right) [120].

Micro- and nanocellulose have been incorporated into PLA for fabricating 3D printing filament and composites. Microcellulose/PLA 3D printing filament has been produced using solvent casting followed by twin-screw extrusion of PLA with the addition of a small amount of microcellulose (<5 wt %) [121]. However, this process results in higher moisture uptake of the microcellulose/PLA filament. Additionally, it has a potential risk of forming bubbles in the filament as a result of the addition of microcellulose, although this risk can be reduced by optimizing the extrusion parameters. Recently, Tekinalp et al. used a similar process to make CNF/PLA filament for 3D printing [122]. The mechanical properties of the 3D printed composites were compared with those made from the traditional compression molding process. At 30 wt % CNF content, the 3D printed sample showed a significantly higher elastic modulus of 7.12 GPa and storage modulus of 1.72 GPa than the compression-molded sample of 6.57 and 0.9 GPa, respectively. The authors proposed a “microsponges” reinforcing mechanism in which the polymer penetrating the freeze-dried CNF porous network during melt-processing creates mechanical interlocking between the CNF and matrix. The higher modulus of 3D printed samples was explained by the higher crystallinity due to an extended temperature being maintained during the printing process, and the possible orientation of CNF bundles along with the printing direction. Although the tensile strength of the printed sample had a remarkable increase up to 80% compared with neat PLA, the mechanical performance of printed composites was inferior to the composites made from compression. This was attributed to the presence of porosity/defects in the printed sample, which can be improved by optimizing the printing conditions (e.g., increased bed temperature for good fusion and lower extrusion temperature for pore reduction).

A commercially available FDM 3D printer was modified with two separate filament supplying paths for polymer matrix and reinforcing fiber, respectively (**Figure 8b**), allowing the production of continuous NFRCs by in-nozzle impregnation [119]. Composites with ~6 vol % jute yarn or carbon fiber were successfully fabricated. The printed jute/PLA had a 134% higher tensile strength and 157% higher modulus than the neat PLA, while the carbon/PLA composite showed much more improvement, 435% and 599%, respectively. Comparable and even superior tensile properties of 3D printed composites can be achieved in comparison with the values of most non-fabric fiber–reinforced PLA (as shown in **Figure 4)**. With only 6.1 vol % of natural fiber, the tensile properties of this printed jute/PLA composite were close to the 20 wt % glass fiber–reinforced PLA [44], even though the performance of the printed jute/PLA composite was thought to be not maximized because the jute yarn was fed into the nozzle with no tension. The fibers without tension could lead to a nonuniform configuration of the twisted jute yarn fiber, resulting in weak points in the printed specimen. Overall, this technique enables direct 3D printing of continuous fiber composites that will substantially expand the methodology of 3D printing for NFRCs with high mechanical performance.

Hinchcliffe et al. demonstrated a technique combining 3D printing and fiber prestressing for high-performance NFRCs [123], which can partially address the lack-of-tension disadvantage observed in the above continuous-fiber composite fabrication. The 3D printed PLA specimen was first prepared with and without post-tensioning ducts. Then, flax and jute fiber strands were threaded through the ducts, stressed to prescribed levels of pretension, and affixed into place using anchors. The measured tensile and flexural properties were transformed into specific mechanical values (e.g., tensile or flexural properties to weight ratio). The results indicated that the reduction in material quantities from ducts showed superior specific mechanical properties. Prestressing of flax fibers threaded in the duct further increased the specific tensile strength and modulus compared with the no-duct PLA specimen. However, this technique showed several shortcomings, including that it (1) is only suitable for high-density matrices (e.g., PLA and HDPE), (2) requires consistent fiber mechanical properties (e.g., fiber strands rather than single fiber), (3) needs complicated designing of duct, and (4) has fiber prestressing loss from loosening anchorage.

Manipulation of the hygroscopic and anisotropic feature of natural fiber leads creatively to a concept of 4D printing of NFRCs, which refers to the responsive capability of 3D printed materials to actuate under an external stimulus. The moisture adsorption of natural fiber has historically been thought to be one of the limitations of the applications of NFRCs. However, this inherent hygroscopic fiber transformation and anisotropic properties have been recently combined with 3D printing technologies for the development of programmable material and responsive architectures [124]. The hygroscopic property of wood fiber enables the composite's structural transformation in response to environmental changes (e.g., temperature, humidity, light radiation), and the integrated anisotropic cell wall structure and 3D printing patterns allow precise control over the directions and degrees of deformation (i.e., curling). Correa et al. demonstrated that self-transformable composites can be fabricated using wood fiber and acrylonitrile butadiene styrene or nylon fiber [124]. After exposure to moisture by water submersion or water vapor, the 3D printed wood composites started curling in a programmed way based on the designed architecture of the composites and the controlled climatic conditions. This curling actuation behavior was found to be fully reversible for more than 30 cycles without any visible defects or changes in curling range. Another study of wood fiber biocomposites found that the printing directions (i.e., 0° and 90°) and printing width significantly influenced the hygromorphic dynamic of the biocomposite because of their differential micro-porous structure [125]. Manipulation of the differential composite microstructure and the anisotropic hygro-elastic and swelling properties enabled a controlled self-actuating functionality of the printed bilayer wood fiber biocomposites (**Figure 9**). The printed bilayer structure was obtained by tailoring the printing directions consisting of an active layer with high radial swelling printed at 0° and a passive layer with low axial swelling at 90° (**Figure 9a)**. The printed biocomposites actuated in response to the external stimulus: hygromorphic bending after immersion in water (**Figure 9b and 9c**). Because the microstructure associated with varying porosity determines the water update and swelling behavior, it therefore alters the actuation response. However, the transformation process of the biocomposites in this study was not fully reversible. Together, taking advantage of the inherently hygroscopic and anisotropic properties of natural fiber and the multi-material and tunable microstructure with 3D printing technologies, high-performance NFRCs with programmable moisture-actuated functionality can be generated.

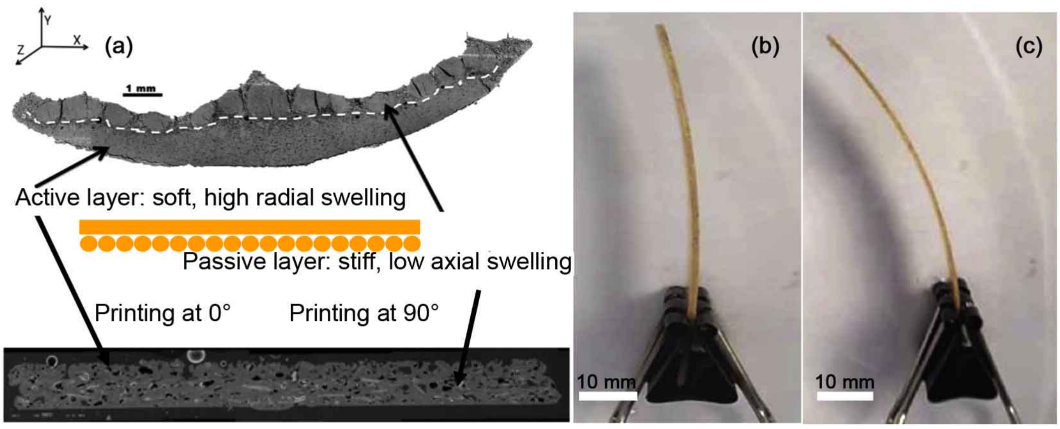

Figure 9. 3D printed hygromorph bilayer wood fiber/PLA/PHA composites. (a) Principle of hygromorph biocomposite design based on the bioinspiration paradigm. Example of hygromorph biocomposite made by FDM (b) before and (c) after immersion. Reproduced from [125] with permission from Elsevier.

4.6 Exploration of new natural fiber sources

A great variety of new plant-based natural fibers and their reinforcing performance for polymer composites have been explored (**Table 4**). Comprehensive characterization of these newly emerged fibers has shown feasible physical, mechanical, thermal, and morphological features as alternative reinforcing natural fibers. A common theme involved in the performance of NFRCs is the chemical, structural, and morphological characteristics of natural fiber. In general, crystalline cellulose content and its microfibril alignment with the fiber axis are positively correlated with the tensile properties of the fiber, whereas the hemicellulose and lignin content correspond to low-stress values and increased moisture absorption and failure strain [126-128]. However, a few other studies have observed positive effects for lignin-containing fibers on the photochemical aging and weathering withstanding of certain NFRCs testing [129-131].

(Table 4. Here)

5. Life cycle assessment

Natural fiber composites may have lower environmental impacts than glass- or carbon-fiber composites if their production impacts are sufficiently low or if they improve the environmental performance of the product.

An LCA concluded that NFRCs can provide significant energy savings and greenhouse gas reduction when used to reduce the weight of aircraft, trucks, automobiles, and other vehicles. These energy savings from light-weighting are larger than the impacts of fiber production. Even so, the impact of natural fiber production is an area that would benefit from further research because agricultural processes used to produce fibers can have significant impacts [156]. Another LCA study of comparing a 3 kg 30% glass-fiber composite component with a 30% cellulose-fiber component and a

40% kenaf-fiber component in automotive applications found that the environmental benefits of NFRCs are primarily because NFRCs are lighter than glass-fiber composites and therefore increase the energy efficiency of automobiles [157]. Across the fibers, the 3-kg part had a life-cycle energy consumption of approximately 20 MJ for fiber production, 150 MJ for resin production, 80 MJ for part manufacturing, 20 MJ for sourcing, 1,200 MJ for use, and 1 MJ for end of life, and 1,450 MJ for total life-cycle energy consumption. Life-cycle energy consumption is largest for the use-phase because in this application, the composite is part of an automobile, so the use-phase energy consumption is proportional to the weight of the component. The cellulose- and kenaf-fiber components weigh less than the glass-fiber component, so the use-phase energy is lower. All the options had approximately the same life-cycle energy for the sum of the fiber production, resin production, part manufacturing, sourcing, and end of life, with the result that the life-cycle energy differences came down to the use-phase and the lightest composite had the lowest impact. The kenaf composite and the cellulose composite provided about 7% reduction in life-cycle energy use of the component, compared with the glass-fiber composites, with potential as high as 19% with resizing of the power train.

Landfilling, incineration with energy recovery, and recycling are three end-of-life options that have been considered [158]. Most fiber-reinforced polymers, including both carbon and glass fibers as well as natural fibers, are landfilled. Landfilling of biomass sequesters carbon, providing a greenhouse gas reduction; incineration with energy recovery releases the carbon in the biomass but can partially displace the use of fossil fuels, providing an alternative route to greenhouse gas reduction. The amount of energy recovered from the incineration of NFRCs ranged from 12 to 34 MJ/kg of the composite, which is more than for glass fiber composites, which are around 7.5 MJ/kg

[156]. Hedlund-Åström studied the end of life of polymer composites and found environmental benefits both for recycling of the flax/PP composite and for incineration for energy content if the process would replace coal combustion [159]. The option of making pellets from the waste NFRCs and combusting them for energy recovery was also considered but was more expensive than the direct waste incineration option. The overall implication of LCA of NFRCs, which have been mostly focused on automotive applications, is that the benefits are mainly due to their light weight rather than their natural origin.

Several studies, however, have pointed out that NFRCs can have inferior environmental impacts than alternatives when certain factors are considered. For example, flax is superior in most environmental impact categories compared with glass fiber, except for eutrophication, because of the use of fertilizers for flax production [160]. This is one of the few studies to examine the potential for eutrophication, and it suggests that other fibers produced with fertilizers may have similar impacts. In a cradle-to-manufacturing LCA, Rodriguez et al. suggested that the use of the banana fiber required changes in the composite to achieve the same functional characteristics, in some cases eliminating the production advantage of natural fiber [161]. Additionally, the fiber treatment method played a vital role in the analysis of the environmental impact of NFRCs [162]. The analysis showed that human health, ecosystem, and resource impacts are mostly independent of whether glass or hemp is used as the reinforcing material in both the reinforced PP or PLA composites [163].

Building on the understanding of the design and production of NFRCs, the lifecycle environmental effects of design requirements have been evaluated. A flax-fiber mat with design optimization for stiffness equal to a glass-fiber mat provides energy savings in automotive uses because the flax fiber mat is lighter. In contrast, in an injection-molded application requiring equal stiffness, even an optimized design for the flax-fiber composite has approximately the same mass and very similar impact as the glass fiber counterpart [164]. Further studies that incorporate the advancing knowledge of natural fiber composite manufacturing and design hold promise for illuminating the interplay between improved physical and environmental performance. Life-cycle costing and social impact measures are emerging that have the potential to provide a more holistic sustainability assessment for natural fiber composites [165].

6. NFRC applications

NFRCs has been widely applied in the automotive sector. It has been estimated that 75% of fuel consumption directly relates to vehicle weight, with every 10% reduction in vehicle weight resulting in 6–8% increment in fuel economy and every 100 kg in weight saving in automotive, leading to approximately 20 g/km reduction in CO_2 emission of commonly used powertrains [2]. Weight reduction directly supports the European Commission's and the European Automobiles Manufacturers Association's goal of reducing CO_2 emission [166, 167]. The European Union End-of-Life Vehicle Directive (European Directive 2000/53/EC) has targeted reuse or recycling 95 wt % of vehicles starting in 2015. In the United States, the Corporate Average Fuel Economy standards require that automakers increase the fuel economy of cars and light-duty trucks to 54.5 mpg by 2025 [27]. Similar regulations have been adopted in many other countries to increase the fuel economy and CO_2 reduction [167].

Early efforts directed at using bio-based materials in vehicle parts were contemplated by Ford Motor Company dating back to the 1930s, and there are several highlighted applications of NFRCs in the automotive industry to date [166, 168]. For example, in 2000, Audi launched the A2 midrange car in which the door trim panels were made of flax/sisal-reinforced PU and Mercedes-Benz disclosed a flax/polyester engine and transmission enclosures for the first time for standard exterior components. By 2019, Porsche announced that the 718 Cayman GT4 Clubsport racecar was the world's first vehicle with bodywork built from hemp/flax reinforced composites. In the past decades, European and North American governmental legislations have encouraged the applications of NFRCs in the automotive sector. These "green" composites derived from natural fibers are increasingly and extensively applied in automotive parts by manufacturers and suppliers, such as door panels, seat backs, headliners, trays, dashboards, and interior parts (**Table 5**). Among these abundant NFRCs, flax-, hemp-, kenaf-, and sisal-reinforced thermoplastics have dominated for automotive component applications. All the major car manufacturers such as Audi, BMW, Daimler Chrysler, Ford Motor Company, General Motors, Mercedes-Benz, and Toyota have used bio-based composites in various applications. For Mercedes-Benz, natural fibers have been extensively applied in components of various models (**Figure 10**) [169]. For example, ~20.8 kg of natural fibers (wood, banana, and flax) in more than 20 components are used for the A-class; ~19.8 kg of natural fibers (wood, coconut, and honeycomb cardboard) in more than 21 components for the B-class; ~17 kg of natural fibers (honeycomb cardboard, charcoal coke, sisal, wood, cotton) in more than 27 components for the C-class; ~21 kg of natural fibers (jute, flax, hemp, sisal, and olive coke) in more than 44 components for the E-class; and ~43 kg of natural fibers in more than 27 components for the S-class. BMW has been using NFRCs since the early 1990s in M3, M5, and M7 series models with up to 24 kg of renewable materials being used.

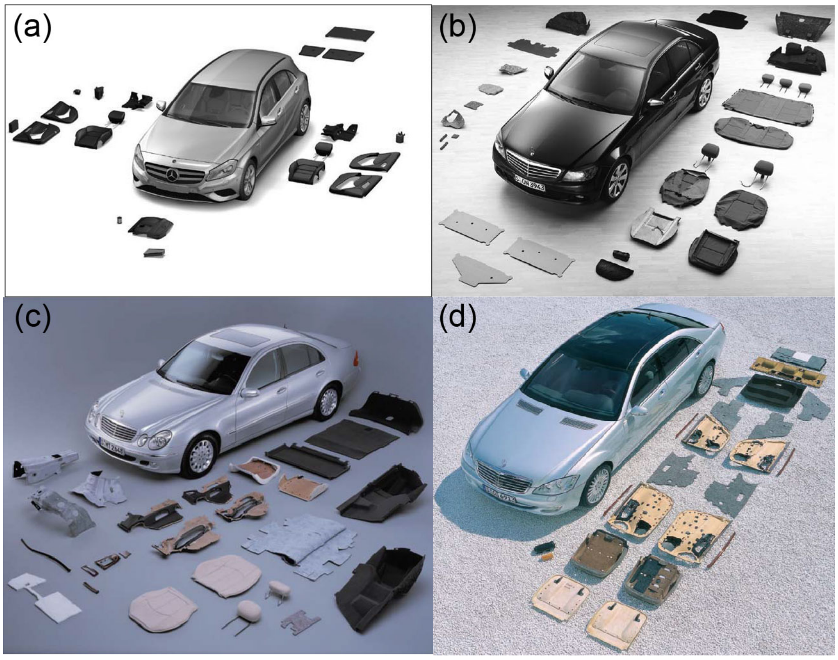

Figure 10. Mercedes-Benz components containing natural fibers (flax, hemp, sisal, wool, and others) of model (a) A-class, (b) C-class, (c) E-class, and (d) S-class. Reproduced from Ref. [17, 166, 169], with permission from John Wiley for (a), (b), and (d), and Springer for (c).

Generally, NFRCs are primarily used in interior automotive parts because of their intrinsic moisture sensitivity and relatively low mechanical properties. Many nonstructural interior parts, such as indoor panels, dashboard, seat backs/fillers/liners, floor mats, package shelves, and storage bins, can include NFRCs. Examples of NFRCs used for exterior components in a production vehicle include flax/polyester composites in the engine and transmission enclosures for sound insulation, abaca-reinforced composites for the spare tire well covers, fender components, spoilers, bumpers, seat frame, and load floors. Recently, new studies on NFRCs, such as flax/Acrodur with impregnated flax tape at short crosslinking times for fast molding parts in vehicles [170], flax/vinyl ester composites for automobile hoods [171], hemp/PP composites resisting weathering for interior and decking components and small auto parts [172, 173], kenaf/epoxy composites for spall liners [81, 82], maple and pine flour–reinforced nylon 6 with enhanced thermal stability for under hood parts [174], palm kernel shell fiber/phenolic resin for brake pads [175], and bamboo/polyurethane composites for sound absorption materials for door panels [176] have expanded the potential automotive applications of NFRCs. Research and development to advance the applications of NFRCs to more structural components are desirable.

(Table 5. Here)

Due to the light weight, low thermal conductivity, and low environmental impacts of renewable materials, NFRCs have been advanced recently for their applications in masonry, soil blending, cementitious materials, thermal insulating materials, furniture, and decks applied in the civil construction, furniture, and architecture (**Table 6**) [179]. However, such composites are generally limited to thin sheet applications with high fiber volume fraction, making them not particularly useful for general construction purposes. Recently, natural fiber composites have been investigated with potentially extensive applications in several other areas, such as sports, clothes, recreation equipment, aerospace, biomedical and pharmaceutical, electrical, packaging, and electromagnetic applications (**Table 6**).

(Table 6. Here)

7. Summary and outlook

NFRCs have well-known appealing characteristics, such as light weight, environmental friendliness, recyclability, and achievable mechanical performance derived from natural fibers. Applying NFRCs, primarily in the automotive sector, can provide weight reduction, energy efficiency, environmental safety, and health benefits. However, the innate heterogeneous characteristics and hydrophilicity of natural fibers raise challenges to the manufacturing and application of NFRCs. The critical issues are fiber quality inconsistency, poor fiber/matrix adhesion, hard fiber dispersion, limited composite manufacturing conditions, and moisture absorption. Exploration of new natural fiber with abundance can diversify fiber sources and reduce material cost. However, consistent quality of natural fibers is always an important feature for a steady performance of NFRCs. Research on the fundamental understanding of the microstructure of the interphase is critical to further improve natural fibers' dispersion and interfacial bonding with polymer matrices and the mechanical properties of NFRCs. Coupling agents and chemical modification can significantly improve the compatibility of natural fiber and polymer matrix, but the interfacial bonding strength seems to play a larger role given the thermo-mechanical orthotropicity of the fiber. In turn, unidirectional fiber, mat fabric stacking, hybridization, and processing revolution strategies can substantially impart the longitudinal performance of fiber into NFRCs. Particularly, natural fibers with superior strength and modulus, such as nano-scaled cellulosic fibers, have emerged and could play a pivotal role as new reinforcing components for polymers. Additionally, the 3D printing of NFRCs with the capability of complex structural design and advanced actuation performance can potentially transform and expand the application of NFRCs significantly in the future. The future market of NFRCs is increasing, and their application, particularly in automotive, civil engineering, sports, and biomedical sectors, is quite promising.

Acknowledgments

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Figures Captions

Figure 1. Relative comparison between natural fiber and glass fiber for composite manufacturing [6, 7]. Insets are SEM photos of (left) alkaline mercerized ramie fiber and (right) E-glass fiber.

Figure 2. Comparison of the reported tensile strengths (σ) and moduli (E) of natural fiber–reinforced PP. Volume fiber fraction is used if the weight loading fraction is not reported. The black dotted line represents the properties of neat PP (σ of 28 MPa and E of 0.69 GPa) [30] and the orange dotted line represents the properties of 20 wt % long glass fiber (GF)–reinforced PP specified for interior applications from Ford Motor Company (σ of 64 MPa and E of 4.1 GPa) [27]. The complete numbers with each reference are included in the supplementary file.

Figure 3. Comparison of the reported tensile strengths (σ) and moduli (E) of natural fiber–reinforced PE. Volume fiber fraction is used if the weight loading fraction is not reported. The black dotted line represents the properties of a neat HDPE (σ of 19.6 MPa and E of 0.74 GPa) [35] and the orange dotted line represents the properties of 20 wt % glass fiber (GF)–reinforced HDPE (σ of 42 MPa and E of 1.4 GPa) [32]. The complete numbers with each reference are included in the supplementary file.

Figure 4. Comparison of the reported tensile strengths (σ) and moduli (E) of natural fiber–reinforced PLA. Volume fiber fraction is used if the weight loading fraction is not reported. The black dotted line represents the properties of neat PLA (σ of 54.36 MPa and E of 3.23 GPa) [37] and the orange dotted line represents the properties of 20 wt % glass fiber (GF)–reinforced PLA (σ of 67.9 MPa and E of 6.7 GPa) [44]. The complete numbers with each reference are included in the supplementary file.

Figure 5. Examples of typically reported methods for treating and modifying natural fibers [53, 54].

Figure 6. Micrographs revealing transcrystalline structure in a sisal fiber/PP composite. (a and b) Polarized optical images and (c and d) SEM images show three types of crystalline structures: spherulite (arrows in a and b), MD-TC (a and c), and HD-TC (b and d). Picture adapted from [68] with permission from Wiley.

Figure 7. SEM micrographs of the fracture surface of composites reinforced with (a) long fiber pellets and (b) short fiber pellets. Reproduced from [111] with permission from Elsevier.

Figure 8. Schematic of FDM process (left) [119] and the FDM-based 3D printer head used to produce continuous fiber–reinforced composite using in-nozzle impregnation (right) [120].

Figure 9. 3D printed hygromorph bilayer wood fiber/PLA/PHA composites. (a) Principle of hygromorph biocomposite design based on the bioinspiration paradigm. Example of hygromorph biocomposite made by FDM (b) before and (c) after immersion. Reproduced from [125] with permission from Elsevier.

Figure 10. Mercedes-Benz components containing natural fibers (flax, hemp, sisal, wool, and others) of model (a) A-class, (b) C-class, (c) E-class, and (d) S-class. Reproduced from Ref. [17, 166, 169], with permission from John Wiley for (a), (b), and (d), and Springer for (c).

Tables

Table 1. Typical properties of commonly studied natural fibers and several synthetic fibers.

Fiber source (or type)	Density (g/cm^3)	Tensile strength (σ/MPa)	Tensile modulus (E/GPa)	Specific modulus (Approx.)	Elongation (%)
Plant-based natural fibers					
Coir [14]	1.2–1.5	95–230	3-6	4	15–51
Cotton [14]	1.5–1.6	287–800	6-13	6	3–10
Flax [15]	1.53	745–1,145	44–61	35	2.07
Hemp [16]	1.48	690	70	47	1.6
Jute [16]	1.3	393–773	27	21	1.5–1.8
Kenaf [14]	1.4–1.5	223–930	15–53	24	1.5–2.7
Ramie [16]	1.5	560	25	17	2.5
Sisal [14]	1.3–1.5	363–700	9–38	17	2.0–7.0
Wood pulp [17]	1.5	1,000	40	27	4.4
Cellulose carbamate [18]	1.5[a]	365	22	15	8
Crystalline cellulose [19]	1.6	7,500–7,700	110–220	103	—
Cordenka rayon [18]	1.8[a]	833	20	11	13
Lyocell [18]	1.4[a]	552	23	16	11
Viscose [18]	1.3[a]	338	11	8	12
Synthetic fibers					
Aramid [17]	1.4	3,000–3,150	63–67	46	3.3–3.7
Basalt [20]	2.7	2,130	93	47	2
Carbon fiber (Hexcel AS4) [21]	1.8	4,330	231	129	1.8
E-glass [14]	2.5–2.6	2,000–3,500	70–76	29	1.8–4.8
S-glass [17]	2.5	4,570	86	34	2.8

—: not reported. [a] linear density (dtex).

Table 2. Selected features of major commercial thermoplastic polymers [23-25].

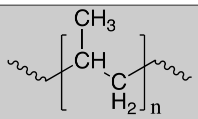
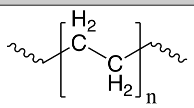
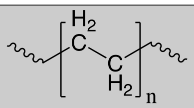
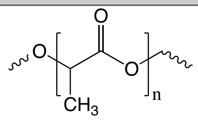
Polymer	Chain structure	T_g (°C)	Density (23/4°C)	Crystallinity (%)	Surface degradation rate on land (μm/year)	Lifespan (year)
PP		~25	0.90	30–50	0.5	10–600
LDPE		~100	0.91–0.93	30–50	1–100	10–600
HDPE		~80	0.96–0.97	80–90	0.6–3.0	10–600
PLA		55–65	1.21–1.25	Amorphous to 90 varying on enantiomers ratio	20–1,000	Months to 100

Table 3. Nanocellulose fiber–reinforced composites and their major properties.

Nanocellulose	Polymer matrix	Tensile strength (MPa)	*E* (GPa)	Other key changes	Ref.
CNC (Kraft pulp)	LDPE	—	0.16→0.27	↓ Thermal degradation ↑ Creep performance	[91]
CNC (Kraft pulp)	PP	—	1.2→1.8	↓ Strain, thermal degradation ↑ Creep performance	[91]
CNC (bleached jute)	Epoxy	44→40	4.6→5.6	↑ Flexural strength and modulus, fracture toughness, E' ↓ T_g, damping factor	[92]
CNC (bleached kenaf)	Polyester	40→48	0.8→0.9	↑ Impact energy, T_g, E'	[93]
CNC (bleached agave, 1%)	PLA	45→37	2.7→3.5	↓ T_c	[94]
CNF (eucalyptus Kraft pulp)	Epoxy	11→14	15% higher	↑ Tribological behavior ↓ T_g	[95, 96]
CNF (bleached agave, 2%)	PLA	45→35	2.7→3.0	↓ T_c	[94]
CNF (bleached softwood)	Epoxy	21→27	0.6→1.3	↑ Strain, impact strength, flexural strength and modulus	[97]
CNF (bleached softwood)	HDPE	22→43	1.1→2.0	↓ Strain	[98]
CNF (bleached carrot)	Cellulose acetate butyrate	—	2.1→3.5	↑ E'	[99]
Microfibrillated cellulose	PP	28→26	1.9→3.1	↑ Flexural strength and modulus ↓ Viscosity	[100]

E: Young's modulus; —: not reported; E': storage modulus T_g: glass transition temperature; T_c: crystallization temperature; → shows the improvement from the neat polymer to the reinforced composite.

Table 4. Properties of recently emerged natural fibers and their reinforced composites.

Fiber source (or type)	Status of fiber (*d*)/ composite (conc., matrix)	Density (g/cm^3)	σ (MPa)	*E* (GPa)	Elong. (%)	Ref.
Agave tequilana	Cooked (329–426 um)	0.88	42–58	2.6–2.9	11–15	[132]
	Injection-molded random fiber (30%, PP)	—	29→32	1.53→2.48	—→4.8	[27]
Althaea officinalis	Solvent and water retted (156–194 µm)	1.2	415	65.4	3.9	[133]
***Areca* fruit fiber**	Water retted (400–500 µm)	0.7–0.8	147–322	1.1–3.2	10–13	[134]
	Random fiber sheet (40%, polyester)	—	37→68	1.5→1.3	—→5.2	[134]
Arundo donax	Stripped (500 µm)	1.2	222	8.6	2.6	[135]
	Hand layup chopped (5%, epoxy)	—	53→74[#]	1.7→3.1[#]	7→—	[136]
Cissus quadrangularis	Water retted (198 µm)	1.2–1.5	207–322	5.6–6.3	5.2	[137]
	Random fiber sheet (40%, polyester)	—	19→36	0.9→2.2	—→4.9	[138]
Conium maculatum	Water retted (—, 200 µm)	—	328	15.8	2.7	[139]
Cortaderia selloana	Random fiber (15%, bio-PE)	—	—→20	—→0.5	—→1.5	[140]
Cynara cardunculus	Water retted (300 µm)	1.6	201	11.6	3	[141]
	Unidirectional (10%, PLA)	1.6	47→50	3.2→4.5	2.5→1.4	[142]
Dichrostachys Cinerea	Water retted	1.2	873	—	—	[143]
Dracaena reflexa	Water retted (176 µm)	0.79	830	46.4	2.9	[144]
Fimbristylis globulosa	Water retted (34 µm)	0.9	452	17.4	—	[145]
Furcraea foetida	Water retted (13 µm)	0.8	~600	6.0–6.5	10.5	[146]
Heteropogon Contortus	Manually plucked	0.6	476	48	1.6	[147]
Ishinosiphon koern	Retted (280 µm)	0.57	614	21	—	[148]
Juncus effusus	Retted (280 µm)	1.1	113	4.4	2.8	[149]
Lygeum spartum	Retted (180–433 µm)	1.5	65–280	4.5–13.3	1.5–3.7	[150]
Manicaria saccifera	Bracts (100 µm)	0.8–1.1	255–391[&]	1.8–2.4	5.0–5.4	[151]
	Natural fabric (40%, PLA)	0.8–1.1	54→68	3.2→4.9	4.9→3.5	[37]
Okra	Random fiber (15%, cornstarch, 90 µm)	—	11→18	0.5→1.6	5.1→2.6	[152]
Pennisetum purpureum	Water retted (150–250 µm)	0.4	73	5.7	1.4	[153]
	Random fiber (25%, polyester)	—	—→16	—→3.0	—	[154]
Ricinus communis	Alkaline retted (20–80 µm)	—	493	20.9	3.0	[155]
	Carded fiber mat, PP (50 wt %)	—	24.1	0.92	—	[155]

d: diameter; σ: tensile strength; *E*: elastic modulus; elong.: elongation at break; —: data not reported; [&] denotes cN/m; [#] denotes flexural strength; → shows the strength changes from neat polymer to reinforced composite.

Table 5. Automobile companies using natural fiber–reinforced composites [6, 166, 168, 169, 177, 178].

Automobile	Natural fibers	Applications
Audi	Flax, sisal	Seat back, side and back door panel, boot liners, hat rack, spare tire liners
BMW	Flax, sisal, wood, cotton, bast	Door panels, headliner panels, boot liners, seat backs, noise insulation panels, molded foot well liners, bumpers, fender liners, shields
Brazilian trucks	Jute coir	Trim parts, seat cushions
Chevrolet Impala	Flax	Trim panels
Citroen	Vegetable, wood	Interior door paneling, parcel shelves, boot liners, mudguards
Daimler-Benz	—	Door panels, windshields, dashboards, business tables, pillar cover panels, glove boxes, instrumental panel support, insulation, molding rods/apertures, seat backrest panels, trunk panels, seat surface/backrests, internal engine covers, engine insulation, sun visors, bumpers, wheel boxes, roof covers
Daimler Chrysler	Coir, cotton, flax, hemp, sisal, abaca	Door panels, engine and transmission covers, pillar cover panels, windshields, dashboards, business tables
Fiat	Bamboo	Door panels
Ford	Kenaf, hemp, rice hulls, tomato, wheat straw	Floor trays, door panels, B-pillars, boot liners, wiring brackets, storage, front grills
General Motors	Flax, hemp, kenaf	Seat backs, inner door panels, cargo area floors
Honda	—	Cargo area
Lotus	Hemp, sisal, kenaf	Body panels, spoilers, seats, interior carpets
Mercedes-Benz	Hemp, sisal, flax, cotton, banana	Inner door panels, door panels, rear panel shelves, engine encapsulation, trunk covers
	Abaca, flax, sisal, wood	Door panels, engine encapsulation, spare wheel pan covers, door liners, seat backrests, parcel shelves, trunk covers
Mitsubishi	Bamboo, flax, hemp, cotton	Cargo area floors, door panels, instrumental panels, floor mats
Opel	—	Instrumental panels, headliner panels, door panels, pillar cover panels
Peugeot	—	Front and rear door panels, seat backs, parcel shelves
Renault	—	Rear parcel shelves
Rover	—	Insulation, rear storage shelves/panels
Saturn	—	Package trays and door panels
Toyota	Kenaf, sugar cane, sweet potato	Door panels, seat backs, floor mats, spare tire covers, package shelves
	Kenaf	Spare tire covers
Vauxhall	—	Headliner panels, interior door panels, pillar cover panels, and instrument panels
Volkswagen	Flax, sisal	Door panels, seat backs, boot-lid finish panels, boot liners
Volvo	Hemp, jute, rapeseed, wood	Seat padding, natural foams, cargo floor trays, dashboards, ceilings

—: not reported.

Table 6. Applications of natural fibers composites in several fields.

Fibers	Other major compositions	Potential applications	Ref.
Civil Construction			
Banana	Soil, cement	Compressed earth block	[180]
Curaua, jute, sisal, ramie, pineapple	Cement, metakaolin, fly ash	Cementitious materials	[181, 182]
Flax, jute, sisal, hemp, coir, oil palm	Polyester, epoxy, mortar	Masonry	[183-189]
Jute	Glass fiber, vinyl ester	Deck panel	[190]
Kenaf	Plaster of Paris	Ceiling	[191]
Wheat straw, corn husk,	—	Thermal insulation materials	[192]
Wood, cellulose, cork	—	Thermal insulation materials	[193]
Furniture and Architecture			
Lignocellulose, straw	HDPE	Plus lounge furniture	[194]
Hemp	Epoxy	Hemp chair furniture	[194]
Hemp, flax	Polyethersulfone	Ignot bio- and Polycal acoustic panel	[194]
Lignocellulose	Thermoplastic, thermoset resin	BioMat research pavilion	[194]
Sports and Clothes			
Cotton, hemp, jute, bamboo, sugarcane bagasse, coconut, banana, raffia	Polyester	Footwear	[195]
Curaua	Aluminum/graphene oxide, epoxy	Ballistic armor materials	[196, 197]
Flax, hemp	Epoxy	Racing bicycle	[198]
Flax	Carbon fiber, epoxy	Bicycle frame	[199]
Jute	Polyester	Winter over coat	[200]
Jute, sisal, coconut, areca, banana	Epoxy	Helmet shell	[201]
Kenaf	Kevlar, epoxy	Ballistic armor materials	[202, 203]
Kenaf	PLA	Mobile phone casing	[198]
Kenaf, pineapple, mengkuang	Epoxy	Recurve bow	[204]
Oil palm empty fruit bunch	Epoxy	Sports utility	[205]
Aerospace			
Hemp	Epoxy	Electronic racks for helicopter	[206]
Ramie	Matrix	Aircraft wing boxes	[207]
Kenaf	Glass fiber, matrix	Aircraft materials	[207]
Biomedical and Pharmaceutical			
Cotton	Epoxy	Medical imaging	[208]
Cotton, sugarcane	PLA, starch	Drugs, antibiotics	[209]
Coconut, sugarcane	Oregano oil, poly(3-hydroxybutyrate-co-3-hydroxyvalerate), starch	Antimicrobial, antibiotics	[209]
Flax, ramie	Epoxy	Bone grafting, orthopedic implants	[210]
Hemp, sisal, coir	Polycaprolactone	Orthoses materials	[211]
Jute	PP	Enzyme	[209]
Jute, sugarcane, flax, cotton, kenaf, bamboo	Plant extracts, polyester, PP, chitosan	Biomedical nanoparticles, antibiotics	[209]
Sisal	Poly(methyl methacrylate)	Drug delivery	[209]
Others			
Bamboo	PLA	Packaging	[212]
Alpha, banana, bamboo, cotton, flax, jute, kenaf, oil palm, sisal	Polyester, PP, PE, PLA, epoxy, rubber, polyurethane, wheat gluten	Dielectric materials	[213, 214]
Flax	Carbon nanotube	Electrodes	[215]
Flax, jute, coir, sisal	Glass fiber, epoxy	Wind turbine blades	[216]
Flax, seagrass	Glass fiber, polyhydroxyalkanoates, epoxy	Marine materials	[217, 218]

Jute	Glass fiber, polyester	Solar parabolic trough collector	[219]
Jute, flax, kenaf, hemp	Aluminum, epoxy, polyester	Electromagnetic interference shields	[220, 221]
Wood cellulose	Cobalt sulfide, carbon fibers	Battery	[222]

—: not reported.